

# Blick - Project Documentation

**Product concept, functional specification, verified data sources, architecture, privacy, and delivery scope**

**Document status:** Android-first MVP specification — see "Current implementation status" immediately below for what already exists in this repository versus what the rest of this document specifies as still planned.

**Updated:** 2 August 2026

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## Current implementation status (as of 2 August 2026)

This document specifies the full intended product. Most of the sections below describe that end-state design in the present tense, as a specification does — they are **not** a claim that all of it already exists. This section is the authoritative summary of what is actually built today; where the two disagree, this section wins.

**Validation status of this update, stated plainly up front:** a complete local run — `testDebugUnitTest`, `lintDebug`, `assembleDebug`, and `connectedDebugAndroidTest` on a physical Lenovo TB350FU (Android 14) — has now passed in full: all 425 JVM `@Test` functions and all 24 instrumented `@Test` functions, `lintDebug` with 0 errors (43 warnings: two expected, already-guarded `InlinedApi` findings — the API-36 `ACTION_APP_NOTIFICATION_PROMOTION_SETTINGS` deep-link and the API-33 `POST_NOTIFICATIONS` permission constant — plus four expected `UnusedAttribute` findings on the widget provider XML's Android-12+-only sizing attributes, kept alongside their legacy fallbacks deliberately), and a working debug APK, with the ongoing-notification loop, routine details live-preview, and full routine management additionally exercised manually on that same device. **The home-screen widget's placement, resizing, live updates, and Stop-action behavior have since been manually confirmed on that same device, including its "Design 1" visual redesign — a colored line-number badge, a large countdown, and a live/scheduled/cancelled status row** — see `android/README.md`'s Full verification pass section for the full account, including the one compact-layout sub-case this device's launcher grid couldn't directly exercise. **Disruptions are now integrated into the Android client** (a Routine Details section plus the notification's expanded view) and were verified on the same device against the real deployed backend for the no-disruption and fetch-failure cases, including live confirmation that a disruptions failure never affects departures — but not for a disruption actually being present, since none was active on the tested line at verification time; see "Disruptions integration, verified end to end on-device" in `android/README.md` for the full account of what was and wasn't observed directly.

Getting there took three work sessions of source beyond the earlier 193-JVM-test baseline. First: the FAB restoration, the Routine Details 30-second auto-refresh, the production notification-permission flow, and the `WorkManagerRoutineScheduler` / `RoutineActiveWindowWorker` scheduling-and-notification-loop implementation, plus a corrective audit fixing a device-local-timezone scheduling bug, sharing one notification-availability check across the notifier/worker/UI, making the Routine Details notification-status hint refresh on every lifecycle resume, and replacing the one-routine FAB's "open then block" flow with an in-place explanation dialog (together, 65 new JVM `@Test` functions and 3 new instrumented ones, reaching 258 JVM / 12 instrumented — all still unrun at that point). Then a verification session actually compiled and ran all of it for the first time, which surfaced and fixed three genuine test-suite issues (an invalid Compose-test-API import, a Robolectric runtime-permission default the tests hadn't accounted for, and a `WorkManager` test-timing assumption that could never hold) — and, more importantly, a real production bug no source review had caught: `RoutineActiveWindowWorker` (a `@HiltWorker`) was never actually receiving its injected dependencies, because the `androidx.hilt:hilt-compiler` annotation processor (distinct from `com.google.dagger:hilt-android-compiler`, already used here for `@HiltViewModel`) had never been added — so `WorkManager` silently fell back to its plain reflection-based factory and crashed on the worker's real constructor every time it was due to run, meaning **enabling a routine never produced the automatic background notification at all**, with nothing surfacing that failure anywhere a user could see it. Fixed by adding the missing processor (see `android/README.md`'s Build section for the full account). A further session added a

dedicated `BOOT_COMPLETED` receiver and durable (Room-backed) stale-snapshot storage across process death, with their own tests, reaching a 260 JVM / 21 instrumented total. Most recently, the ongoing notification was reworked to request promotion to Android 16's Live Update surface (see "Requesting a promoted Live Update" under "Active-window scheduling and the 30-second notification loop" below), a Stop action was added to it (see the same section), and a pre-existing device-timezone bug in the routine details screen's "pause today"/"resume today" controls was fixed to agree with the worker's own device-zone definition of "today" — ten further JVM `@Test` functions across these three changes reached 270 JVM / 21 instrumented. Most recently still, the active-window worker's notification-availability check — previously only performed once before the loop began — was extended to re-run on every loop tick, so the loop now stops fetching departures and burning battery/network for the rest of the window as soon as the user disables notifications partway through, rather than continuing until the window's own end; this added one further JVM `@Test` function, reaching 271 JVM / 21 instrumented. A further session then: corrected the debug notification section's promotion-status wording to state plainly that `canPostPromotedNotifications()` is an eligibility check, not confirmation that any OEM surface actually rendered a card; added a deep-link from the routine details screen to Android's own per-app Live Update settings (`Settings.ACTION_APP_NOTIFICATION_PROMOTION_SETTINGS`); added a simple About screen carrying the required Trafiklab.se attribution, reachable from the routine list's top app bar; and fixed routine setup so a route needs only to run at least once within roughly the next 20 hours to be selectable, not at the exact moment of setup, by requesting SL Transport's forecast window at Blick's own tested 20-hour cap instead of the live-display default (see "Initial setup" under §7 below). This added 4 further JVM `@Test` functions and 3 further instrumented ones, reaching 275 JVM / 24 instrumented. A correction pass then fixed a real bug that session had introduced — the Live Update settings link was shown (and tapped, doing nothing) on every Android version, not only 16+ where its target settings screen exists — by gating it behind `Build.VERSION.SDK_INT >= Build.VERSION_CODES.BAKLAVA`, and fixed a handful of documentation inaccuracies alongside it: an incorrectly named "resolvability check" that the code never actually performed (it only ever catches `ActivityNotFoundException`), the `forecast` cap described as an officially "supported maximum" rather than Blick's own empirically observed value, a stale backend test count, an overclaim that disruptions appear in the notification today, and a direction-discovery empty-state message that still said "right now" despite the 20-hour window. This added 3 further JVM `@Test` functions, reaching 278 JVM / 24 instrumented. A second correction pass then fixed the settings link's remaining silent-failure case — tapping it on an Android 16+ OEM build without that specific Settings screen still did nothing — by falling back to the ordinary per-app notification settings screen instead; corrected a stale "will actually promote" comment in `RoutineDetailsViewModel` to "is currently eligible for promotion"; made the framework diagram explicitly show "request promotion → OEM decides placement → ordinary notification is the fallback"; corrected the remaining "resolvability check"/"supported maximum" overclaims and a disruptions overclaim in `docs/api-contract.md`; and added an explicit backup/transfer decision (`xml/data_extraction_rules.xml`, `xml/full_backup_content.xml`) that a saved routine must never transfer or restore onto a new device by any mechanism, and a monochrome launcher-icon layer with `android:roundIcon`. This added 2 further JVM `@Test` functions, reaching 280 JVM / 24 instrumented. A further session then implemented the home-screen widget itself: `RoutineWidgetMapper` (the widget's exact counterpart to `RoutineNotificationMapper`, reusing the same `LiveDeparturesState` input, `countdownMinutes` recomputation, and expired-departure filter), `RoutineWidgetUpdater` (driven only by `RoutineActiveWindowWorker`'s existing ~30-second loop and every routine-lifecycle mutation site — no second worker, timer, foreground service, or departure engine), and `BlickRoutineWidget` (Jetpack Glance, `updatePeriodMillis="0"` so Android's own widget-update scheduler is never used). This added 53 further JVM `@Test` functions with no further instrumented ones, reaching 333 JVM / 24 instrumented. A final re-audit session then fixed several real lifecycle/visual gaps in that first widget implementation: `StopRoutineNotificationAction` never reconciled the widget; several worker exit paths (missing/deleted/disabled routine, an elapsed window, notifications unavailable at startup, a handled failure) left it stale instead of clearing or reconciling it; a window that was active but had notifications blocked left the widget on `Loading` forever instead of saying so; a routine's own rescheduling after such a case could ask `WorkManagerRoutineScheduler` to immediately re-enqueue the SAME still-open occurrence with a zero delay, risking a tight retry loop; a freshly-placed widget instance defaulted to "No

active commute." until the next lifecycle event instead of deriving its real current state immediately; `GlanceAppWidget`'s actual default `SizeMode.Single` (not `Exact`, as first assumed) meant it never responded to resizing; and its default un-themed text color would have been unreadable against a themed background. All fixed — see "Home-screen widget" below for the design of each — and covered by 8 further JVM `@Test` functions, reaching 341 JVM / 24 instrumented. A "Design 1" visual redesign session then replaced the simple departure-row list with a colored line-number badge (`LineBadgeColorMapping`, mapping Stockholm's own per-line-family colors — Pendeltåg pink, Metro blue/red/green lines — see "Home-screen widget" below for the exact mapping), a large next-departure countdown with the station → direction and following departure beside it, and a live/scheduled/cancelled status row, with font sizes scaling through width-driven responsive tiers rather than one fixed size (the reference mock was captured on a tablet-sized placement, whose grid cells are physically much larger than an ordinary phone's). This added 23 further JVM `@Test` functions with no further instrumented ones, reaching 364 JVM / 24 instrumented. A further session then wired the previously-built-but-unused `DisruptionRepository` into the Android client for the first time: a shared `DisruptionCache` (60-second TTL, in-flight de-duplication so the active-window worker and the Routine Details screen's own auto-refresh never trigger more than one upstream-bound fetch per filter per minute between them), a `GetDisruptionsUseCase/DisruptionsState` mirroring the departures engine's own Loading/empty-is-not-a-failure/ failure split, a new "Disruptions" section on the Routine Details screen (loading, no-disruptions, unavailable, and a priority-ordered list), and the ongoing notification's expanded view showing the single highest-priority disruption's header and details, appended after the existing departure/last-checked lines without ever touching the collapsed countdown, departure text, Stop action, or Live Update request — see "Disruptions" under §8 and §9 below for the full design. Also fixed a backend filtering bug found in the same phase of work: a line-only disruption (one with no station scope at all) was being unconditionally excluded regardless of the requested line, because an empty `scope.stop_areas` array made the `siteld` check always fail; it is now included when both the requested line and transport mode match the deviation's own scope, while a deviation scoped to an unrelated station continues to be excluded exactly as before. This added 61 further JVM `@Test` functions with no further instrumented ones, bringing the fully verified total to 425 JVM / 24 instrumented, stated above.

### Implemented today (Android client + backend):

- Live SL stop search and transport-mode/line/direction discovery during routine setup.
- The routine-creation wizard, backed by Room persistence.
- A foreground routine details/live-preview screen showing **up to two departures total** (not three — see the corrected departure-count language later in this document), which now fetches immediately when the screen becomes active and again automatically about every 30 seconds while it stays visible (lifecycle-`STARTED`), stopping when the screen is no longer visible and restarting with an immediate fetch when it becomes visible again. A manual "Refresh" action remains available alongside this automatic refresh.
- Routine editing (the same wizard, in an edit mode), enable/disable, pause-today / resume-today (with automatic cleanup of an expired pause), and deletion behind an in-screen Material 3 confirmation dialog.
- An always-visible "Add routine" floating action button on the routine list screen, shown whether the list is empty or already has a saved routine, opening the same creation flow either way.
- A first-beta one-routine limit enforced at the application/UI level: the "Add routine" FAB always stays visible, but with one routine already saved, tapping it shows an in-place dialog explaining the beta limit and pointing at editing/deleting the existing routine, rather than opening a creation flow that could never actually save. `RoutineCreateViewModel.oneRoutineLimitReached` still blocks saving a second routine directly, kept as defence in depth for any other entry point into that screen (e.g. a deep link), but the list screen itself never navigates there once a routine exists.
- Loading, live, no-departures, offline, stale, and unavailable states for the departures section — including a fix ensuring that editing a routine to a different site/line/direction/transport-mode, followed by that new

configuration's first fetch failing, can never surface the *previous* configuration's departures mislabelled as "stale" data for the new one. The client's retained last-successful snapshot is now scoped to the exact departure identity (site, line, direction, transport mode) that produced it, and is discarded rather than reused across an identity change.

- The backend's full contract, request validation, upstream normalization, and caching logic (265 passing automated tests as of this update).
- Disruptions: the Routine Details screen shows a dedicated section (loading, no-disruptions, unavailable, and a priority-ordered list of header/details) for the routine's own site/line/transport-mode, and the ongoing notification's expanded view shows the single highest-priority disruption's header and details, appended after the existing departure/last-checked lines — never replacing or hiding the collapsed countdown, departure text, Stop action, or Live Update request. Both are backed by the same shared, TTL-capped `DisruptionCache` so the background worker and the details screen's own auto-refresh never trigger more than one upstream-bound disruptions fetch per filter per minute between them; a disruptions failure is fully isolated and never stops, delays, or replaces a departures update. See "Disruptions" under §8 and §9 below.
- The ongoing-notification foundation (a pure mapper from a routine + the live-departures engine's state + the current time to a notification presentation model, recomputing each departure's countdown rather than trusting a cached value; a real `NotificationCompat`-based builder with one quiet `IMPORTANCE_LOW` channel, one stable notification id reused on every update, a `BigTextStyle` body listing both departures plus a short critical-text countdown for the soonest one, distinct wording for the live/stale/no-departures/offline/unavailable/loading states; and tap-to-reopen-the-routine navigation) **now runs automatically**, not only through the debug-only manual trigger (which remains, in debug builds only, as a development aid): see "Active-window scheduling and the 30-second notification loop" below. The builder also requests promotion (`setRequestPromotedOngoing(true)`) to Android 16's Live Update surface (Samsung's Now Bar where supported) on every post; a `PromotedNotificationChecker` wraps `NotificationManagerCompat.canPostPromotedNotifications()` so the Routine Details debug section can surface whether promotion is actually available on the current device, but the same notification is posted either way — an unsupported or promotion-disabled device automatically falls back to a normal ongoing notification with no separate code path. The notification also always carries a Stop action ("Stop/Unpin" the current window early — same effect as "pause for today" — see "Requesting a promoted Live Update" below).
- Production runtime notification-permission onboarding: a rationale dialog gated behind `AppSettingsDataStore.hasSeenNotificationRationale`, shown at the point the user enables or saves a routine that is meant to show notifications, never repeated after being dismissed once, with a route to the system notification-settings screen when the app's notifications or the Blick channel are disabled. The Routine Details screen also shows a read-only notification-status line (active vs. disabled) computed via the same shared `NotificationAvailabilityChecker` the notifier and the scheduling worker both use, so all three always agree — it never claims automatic delivery is active when app notifications or the Blick channel are actually off — and that status line now re-checks on every lifecycle resume (e.g. returning from system notification settings), not just once when the screen first loads.
- Active-window scheduling and the 30-second notification loop: saving, editing, enabling, disabling, pausing, resuming, or deleting a routine schedules, replaces, or cancels a `WorkManager` one-time work item for that routine's next active window (unique per-routine work name, so a change can never leave stale scheduled work behind), computed against the device's own local time zone (resolved fresh on every call, so a live device timezone change is picked up without any extra wiring, and a routine's configured wall-clock times are never misinterpreted against UTC or any other fixed zone). When that window starts, a `CoroutineWorker` first checks shared notification availability; only if notifications are actually available does it enter foreground execution, fetch departures immediately, and post or silently update one ongoing,

`setOnlyAlertOnce(true)` notification (the same notification id on every update — never a new card) that requests promotion to a lock-screen Live Update, waiting about 30 seconds and repeating until the routine's configured end time. If notifications are unavailable (permission missing, app disabled, or the Blink channel disabled), the worker never enters foreground execution, never starts the loop, and never reports delivery as active — it simply reschedules the next eligible occurrence and exits. On normal completion, a handled failure, or unavailability, the worker re-reads the routine and reschedules its next eligible occurrence (unless the routine has since been deleted or disabled); a genuine cancellation from an edit, delete, or replacement is never treated as a completion and never resurrects the cancelled work. See "Active-window scheduling and the 30-second notification loop" in the architecture section below for exactly how this is built and what its real-world timing limitations are.

#### Now implemented, beyond the original plan for this milestone:

- A dedicated `BOOT_COMPLETED` receiver (`scheduling/BootCompletedReceiver`), alongside `BlickApplication.onCreate()`'s existing process-start reconciliation and the runtime-registered `ACTION_TIMEZONE_CHANGED` receiver — all three call the same `RoutineScheduleReconciler.reconcileAll()`, which only ever enqueues WorkManager work; it never starts a foreground service directly from the `BOOT_COMPLETED` broadcast itself, so the "no `dataSync` foreground service directly from `BOOT_COMPLETED`" restriction on modern Android (see the architecture section) is still respected — WorkManager decides when the actual foreground worker later runs.
- Persistent stale-data storage: the last successful departure snapshot used for the `Stale` fallback is now Room-backed (`data/local/room/StaleSnapshotEntity.kt`), keyed by routine id and scoped to the exact site/line/direction/mode that produced it, and shared between the routine details screen and the background worker — it survives process death, unlike the previous in-memory-only session scope.
- A deep-link to Android's own per-app Live Update settings (`Settings.ACTION_APP_NOTIFICATION_PROMOTION_SETTINGS`): the routine details screen now offers this when notifications are available but promotion currently isn't eligible, shown only on Android 16+ (promotion eligibility is unconditionally `false` below that, and the settings screen itself doesn't exist there). On an OEM build that omits the screen even on Android 16+, `launchLiveUpdateSettings` catches the resulting `ActivityNotFoundException` and falls back to the ordinary per-app notification settings screen rather than leaving the tap do nothing. This is Android's own general per-app control, distinct from and unable to affect Samsung's separate "Live notifications for all apps" developer-only gate — see "Requesting a promoted Live Update" below.
- A simple About screen (`ui/screens/about/AboutScreen`, reached via an info icon in the routine list's top app bar) satisfying MVP requirement 17: app name, version, the required Trafiklab.se attribution text with a link to Trafiklab.se, and a non-affiliation disclaimer.
- An explicit backup/transfer decision to keep Blink simple: `AndroidManifest.xml` declares `android:dataExtractionRules` and `android:fullBackupContent` (see the referenced XML files' own doc comments), and a saved routine must never transfer or restore onto a new device by any mechanism. `android:allowBackup="false"` already disables cloud backup and adb backup everywhere, but Android's own docs note it doesn't reliably disable Android 12+ device-to-device transfer on every OEM; both XML files explicitly exclude every backup/transfer domain (`root`, `file`, `database`, `sharedpref`, `external` — treated as separate, non-overlapping domains by Android's schema, not a hierarchy under `root`) from both cloud backup and device transfer, so the Room database and DataStore preferences are covered either way.
- A monochrome layer (`ic_launcher_monochrome.xml`) on the adaptive launcher icon for Android 13+ themed-icon support, and `android:roundIcon` in the manifest, now pointing at the `ic_launcher_round` adaptive-icon resource that already existed but was unused.
- A home-screen widget (`widget/BlickRoutineWidget`, built with Jetpack Glance) — see "Home-screen widget" under §8 below for the full functional description. Deliberately reuses every existing piece rather



than building a second departure engine: `RoutineWidgetMapper` mirrors `RoutineNotificationMapper` exactly (same `LiveDeparturesState` input, same `countdownMinutes` recomputation, same expired-departure filtering), and `RoutineWidgetUpdater` is driven by `RoutineActiveWindowWorker`'s existing ~30-second loop, every one of that worker's exit paths that cannot continue an active commute, every routine-lifecycle mutation site (including the notification's own Stop action), and a freshly-placed widget instance's own first render — no second worker, timer, foreground service, or refresh mechanism. Live widget updates depend on notification availability by design: when a window is active but notifications are blocked, the widget shows a dedicated `NotificationsUnavailable` state rather than a misleading placeholder. Android's own widget-update scheduler is explicitly disabled (`updatePeriodMillis="0"`); the widget responds to resizing (`SizeMode.Exact`) without clipping, against a theme-aware readable background. Its placement, resizing, live updates, and Stop-action behavior have been manually confirmed on a real launcher — see [android/README.md](#)'s Full verification pass section for the complete account.

**Not yet implemented** (described in the sections below purely as the plan):

- Exact-time activation — see "Active-window scheduling" below for why this is deliberately best-effort, not exact.
- The home-screen widget does not show disruptions — by design, not an oversight (see "Home-screen widget" below): it reuses the notification/departures pipeline only, and disruptions were deliberately kept out of scope for it in this phase.

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## 1. Project summary

**Blick** is an Android-first application that automatically displays upcoming SL bus, metro, commuter-train, tram, local-rail, and compatible ferry departures during times chosen by the user.

Instead of requiring the user to open a journey-planning application and search for the same stop every morning, Blick places the relevant information in one quiet, updating Android notification. The user selects an SL site, transport mode, line, direction, active weekdays, and a time window. During that window, the application shows the next relevant departures and disruptions affecting the selection.

The first release is a native Android application. A native iPhone client is planned for a later phase and will use the same platform-neutral backend contract. A Smart TV departure display is a possible future feature, not part of the MVP.

The application is intended to be a calm information display. It does not tell users when to leave, predict whether they will catch a departure, or use stressful messages such as "Leave now" or "Probably too late."

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## 2. Problem

Many commuters repeatedly check the same information:

- the same bus stop or station;
- the same line and direction;
- at approximately the same time;
- on the same weekdays.

Existing journey-planning applications usually require several actions: unlock the phone, open an application, search for a stop, station, or journey, and inspect the results. This is unnecessary when the regular commute is already known.

Blick reduces this repeated interaction by showing the information automatically when it is relevant.

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### 3. Proposed solution

The user creates a scheduled commute routine, for example:

| Setting       | Example            |
|---------------|--------------------|
| Site          | Fruängen           |
| Transport     | Metro              |
| Line          | 14                 |
| Direction     | Toward T-Centralen |
| Days          | Monday-Friday      |
| Active period | 07:30-08:00        |

At 07:30, a quiet, ongoing notification appears:

#### **Fruängen to T-Centralen**

Next: 4 min

Then: 11 min / 18 min

The same notification updates during the selected period. When the first departure is no longer relevant, the following departure becomes the next one. At 08:00, the notification disappears and active updating stops.

When there is a relevant disruption, it is added below the departures:

#### **Fruängen to T-Centralen**

Next: 4 min

Then: 11 min / 18 min

Disruption: Delays on line 14 due to a signal fault

If no relevant disruption exists, the disruption area is not shown.

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### 4. Product principles

#### **Calm**

The application presents neutral information without pressure, judgement, repeated alerts, or urgency-based language.

#### **Automatic**

Once a routine has been configured, the user should not need to open the application every day.

#### **Relevant**

Only departures and disruptions connected to the selected site, transport mode, line, and direction should be displayed.

#### **Glanceable**

The most important information must be understandable within a few seconds from the lock screen or notification panel.

## Respectful of attention

The application updates one ongoing notification instead of sending a new notification every few minutes.

## Honest about data

The application distinguishes scheduled information, real-time information, stale information, and unavailable information. It does not invent missing fields or present old data as current.

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## 5. Supported SL transport

The first version is intended to support:

- buses;
- metro;
- commuter trains;
- local and light rail;
- trams;
- SL ferry departures when compatible data is available.

The verified upstream transport-mode values are **BUS**, **METRO**, **TRAM**, **TRAIN**, **SHIP**, **FERRY**, and **TAXI**. The Android UI will expose only modes included in the product scope.

The same routine model applies to each supported mode: the user selects an SL site, a line where relevant, a direction, active days, and an active time window.

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## 6. Target users

The initial target users are regular SL commuters who:

- travel from the same site at a similar time on workdays or study days;
- want to check departures while preparing at home;
- prefer passive information over repeatedly searching in a journey planner;
- want relevant disruption information without unrelated network-wide announcements.

The first product is SL-specific. Support for other public-transport operators is a possible later expansion and is not part of the current architecture contract.

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## 7. Core user experience

### Initial setup

*All eight steps below are implemented today; step 8's rationale dialog is shown once, at the point the user first enables or saves a routine meant to show notifications — see "Current implementation status" above.*

- 1 The user opens the Android application.
- 2 The user searches for and selects an SL site.
- 3 The user selects a supported transport mode.
- 4 The user selects a line and direction.
- 5 The user selects active weekdays.



- 6 The user selects a start and end time.
- 7 The user saves the commute routine.
- 8 The application requests notification permission and explains lock-screen visibility.

Line and direction options are initially discovered from live departures at the selected site, requesting the SL Transport departures endpoint's `forecast` parameter at Blick's own tested cap of 1200 minutes (≈20 hours — an empirically observed value, not a maximum documented or guaranteed by Trafiklab/SL; see [docs/api-contract.md](#)'s departures endpoint entry) rather than the shorter window used for live display, so a route need only run at least once within roughly the next 20 hours to be selectable during setup, not at the exact moment of setup. A route that runs less often than that may still be unavailable during setup. This is a documented MVP limitation. Direction discovery must remain behind an application interface so a more complete source can replace it later without changing the saved-routine model.

## Daily operation

*All 7 steps below are implemented in source and verified on a real device — see "Current implementation status" above and the "Active-window scheduling and the 30-second notification loop" architecture note below. Step 6's disruption fetch was verified live for the no-disruption and fetch-failure cases; a disruption actually being present was verified only through the automated test suite, not by direct observation of a real one rendering on-screen — see "Current implementation status" above for the exact account.*

- 1 The routine becomes active around the configured start time (best-effort, not exact — see the scheduling section below).
- 2 The application requests current normalized departures from the Blick backend.
- 3 It filters by the saved site, line, transport mode, and direction code.
- 4 One ongoing notification appears.
- 5 The notification is silently refreshed about every 30 seconds during the active period, without a repeated sound, vibration, or extra card.
- 6 Relevant disruption information is included when available, as the highest-priority currently-relevant disruption's header and details, appended after the departure lines in the notification's expanded view.
- 7 The notification is removed at the configured end time, and the next eligible occurrence is scheduled.

Separately, and independently of the notification loop above, the Routine Details screen performs its own 30-second automatic refresh purely to keep what's on-screen current while the user is actually looking at that screen — it starts as soon as the screen becomes visible, stops as soon as it isn't, and has no effect on whether the background notification loop is running.

## User controls

The notification or application may offer:

- **Refresh**
- **Pause today**
- **End now**
- **Open routine**

These controls should remain optional and visually secondary.

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## 8. Functional requirements

### Commute routines

The application must allow the user to:

- create a commute routine;
- choose an SL site;
- choose a supported transport mode;
- choose a line and direction;
- choose one or more weekdays;
- set a start and end time;
- enable or disable a routine;
- edit or delete a routine;
- pause a routine for the current day.

The first UI supports one routine. The local database is designed for multiple routines so later expansion does not require replacing the storage model.

## Departure display

During an active routine, the application must:

- retrieve current departures through the versioned backend API;
- display up to two upcoming departures total, when available;
- show the selected site, line, and direction;
- calculate the visible countdown locally;
- automatically remove departures whose effective time has passed;
- update the existing notification rather than creating repeated notifications;
- show when the source response was fetched when data may be stale;
- distinguish real-time and scheduled-only information;
- show cancellations when they can be determined from verified upstream fields;
- handle missing, malformed, or unavailable data safely.

## Disruptions

The application must:

- retrieve detailed SL disruption messages through the backend;
- match disruptions using the same SL site and line namespace as departures;
- select a Swedish message variant when available;
- display relevant disruptions in neutral language;
- avoid unrelated network-wide messages;
- show cancellations, changed routes, station closures, and significant delays when provided;
- hide the disruption section when nothing relevant is reported.

## Notification and lock screen

*The notification foundation (mapper, builder, tap navigation), the production permission-rationale flow, and the automatic active-window scheduling/posting/removal loop are all implemented in source and verified — see "Current implementation status" above for the full local run (JVM unit tests, `lintDebug`, `assembleDebug`, and `connectedDebugAndroidTest`) plus manual exercising on a physical device, covering the ongoing-notification loop, mid-window edits, and full routine management. Promotion to Android 16's Live Update surface has been*

*confirmed possible on a real Samsung Galaxy S23 Ultra, but only after manually enabling a Samsung-specific Developer options flag — Samsung currently blocks third-party Now Bar access for ordinary users on both One UI 8 and 8.5, with no official removal date found, so a regular user on that same real Android 16 device sees no promoted card by default today, through no fault of Blick's own implementation, and this should be treated as device/firmware-dependent rather than a guaranteed feature; see "Requesting a promoted Live Update" below for the full account.*

The application must:

- request Android notification permission;
- use one ongoing notification during the active period;
- permit full departure information on the lock screen when allowed by the user's system settings;
- remove the notification when the routine ends;
- explain that Android and the user's privacy settings control lock-screen visibility.

The intended presentation is not merely an ordinary notification-drawer entry: while a routine is active, the application must request promotion of that ongoing notification to Android 16's Live Update system surface — a prominent, persistent card intended for the lock screen and, where supported, Samsung's Now Bar — rather than treating an unpromoted ongoing notification as the finished feature. Concretely, the application must:

- declare the `android.permission.POST_PROMOTED_NOTIFICATIONS` manifest permission (a normal, non-runtime permission, separate from the runtime notification permission above);
- request promotion on the routine's ongoing notification (`setRequestPromotedOngoing`), which requires an ongoing notification with a title and a system-supported style — no custom `RemoteViews`;
- check whether the current device and user settings actually allow a promoted post (`canPostPromotedNotifications`) and surface that status for verification rather than assuming promotion always succeeds;
- fall back automatically to a standard ongoing notification — the same content, same update cadence, same removal behavior — on devices or configurations where promotion is unsupported or disabled, with no separate code path and no degraded content;
- provide a Stop/Unpin action on the notification itself, stopping the current active window early (the same effect as "pause for today") without requiring the app to be opened.

The exact visual treatment of a promoted Live Update (whether it renders as a card, where exactly it appears, and any OEM-specific eligibility rules) is controlled by Android and, on Samsung devices, One UI — Blick requests promotion but cannot guarantee that every device grants it.

## Home-screen widget

Implemented (`widget/BlickRoutineWidget`, Jetpack Glance — see "Current implementation status" above and `android/README.md`'s Status section for the full account; its placement, resizing, live updates, and Stop-action behavior — including its "Design 1" visual redesign — have been manually confirmed on a real launcher — see `android/README.md`'s Full verification pass section). A header row shows the routine's own pinned line number in a small rounded badge, colored by `LineBadgeColorMapping` (`widget/LineBadgeColorMapping.kt`) per Stockholm's own per-line-family convention — Pendeltåg (commuter rail) lines 40/41/42X/43/43X/44/48 in pink (`#FF49A5`), Metro blue-line 10-11 in blue (`#177BC0`), Metro red-line 13-14 in red (`#EE2D28`), Metro green-line 17-19 in green (`#51BA5B`), bold white badge text on every color, every other mode/line combination in a neutral grey — followed by the destination. Below that: a large, bold next-departure countdown on the left, the station → direction line and the following departure's own smaller countdown on the right (a minute-only countdown recomputed at render time, never a fixed or cached value, reusing the exact same `countdownMinutes` function and expired-departure filter the notification uses), and a colored dot plus "Live"/"Scheduled"/"Cancelled" label underneath reflecting the

next departure's own real-time/cancelled state. The same loading/live/stale/offline/unavailable/no-upcoming-departures states apply, with a cancelled departure represented in the countdown/status area rather than a separate top-level case, plus a widget-only `NotificationsUnavailable` state (no notification counterpart) shown when a window is active but notifications are blocked — live widget updates depend on notification availability by design, since the worker's loop is the widget's only data source, exactly like the notification. Outside any active window it reads exactly "No active commute." Tapping it opens the routine details screen, reusing `MainActivity`'s existing notification-tap navigation contract unchanged. Updated from `RoutineActiveWindowWorker`'s existing ~30-second active- window loop, every one of that worker's exit paths that cannot continue an active commute (missing/deleted/disabled routine, an elapsed window, notifications unavailable at startup or discovered mid-loop, a handled failure before or during foreground execution), every routine-lifecycle mutation (create/edit, enable/disable, pause/resume, delete, the notification's own Stop action, and reboot/timezone/process-start reconciliation), and a freshly-placed widget instance's own first render — no second worker, timer, foreground service, or departure-fetching engine, and Android's own widget-update scheduler is explicitly disabled (`updatePeriodMillis="0"`). Explicitly overrides `SizeMode.Exact` (`GlanceAppWidget`'s own default, `SizeMode.Single`, would not respond to resizing) so the layout adapts to the widget's live size without clipping — badge, countdown, and secondary text scale through four `LocalSize`-width-driven responsive tiers (chosen because the "Design 1" reference mock was captured on a tablet-sized placement, whose grid cells are physically much larger than an ordinary phone's), and the secondary block/status row are dropped below a height threshold — against a theme-aware readable background. All installed widget instances update together; there is no per-instance configuration screen, since the existing first-beta one-routine limit means every instance has only one possible routine to show.

---

## 9. Verified data sources

### SL Transport API

SL Transport is the primary source for:

- the complete list of SL sites;
- site coordinates and child stop-area IDs;
- stable SL line IDs;
- directions and direction codes;
- scheduled and expected departure times;
- stop-area and stop-point information;
- journey and departure states;
- embedded trip and site deviation signals.

The backend uses:

- `GET /v1/sites?expand=true`
- `GET /v1/sites/{siteId}/departures`

The API is currently keyless. Fair-use requirements still apply.

Official documentation: [SL Transport](#)

### SL Deviations API

SL Deviations provides detailed disruption records, including:

- stable deviation-case IDs;

- created and modified timestamps;
- publication windows;
- priority fields used for sorting;
- Swedish and other message variants;
- affected stop areas and lines.

The backend uses:

- `GET /v1/messages`

SL Transport Site IDs have been verified as accepted by the SL Deviations `site` filter. Line IDs and stop-area IDs also align with their corresponding entities in SL Deviations responses. No Trafiklab Stop Lookup conversion is required. The Deviations service is keyless but its documentation asks consumers to request data no more than once per minute.

Official documentation: [SL Deviations](#)

## Excluded upstream APIs

Trafiklab Timetables and Trafiklab Stop Lookup are not used by the SL-only MVP. Removing them avoids incompatible ID namespaces and the absence of a stable numeric line ID in the Timetables route model.

## Licence and attribution

Before public release, the operator must satisfy the current SL API terms, including any required registration and acceptance. The product must visibly state that its transport information is based on data retrieved from Trafiklab.se. When practicable, the attribution should link to Trafiklab.se.

Recommended attribution:

Based on information from Trafiklab.se

The application must not imply endorsement by or affiliation with SL, Trafiklab, or Samtrafiken. The first version uses its own visual identity and plain transport labels rather than protected logos or copied brand styling.

Official terms: [SL API licence](#)

# 10. Technical architecture

## Repository

The project uses one repository with separate areas:

```
blick/
  android/
  backend/
  docs/
```

The Android application and backend have separate build systems and tests. `docs/api-contract.md` holds the detailed machine-facing contract. This project documentation remains the product and architecture overview.

## Native Android application

The Android client uses:

- Kotlin;
- Jetpack Compose and Material 3;

- MVVM with repository interfaces;
- Hilt for dependency injection;
- Room for commute routines;
- Preferences DataStore only for small application settings;
- Retrofit with kotlinx.serialization for the backend API;
- WorkManager (not AlarmManager — see "Active-window scheduling" below for why) behind a `RoutineScheduler` interface, implemented by `WorkManagerRoutineScheduler`;
- Android notification APIs behind a notifier interface.

Pinned SDK values:

```
compileSdk = 36
targetSdk = 36
minSdk = 26
```

`minSdk = 26` is a product support decision. It is not described as a technical requirement for notification channels.

Build toolchain: Android Gradle Plugin 9.2.1 using AGP's built-in Kotlin (Kotlin 2.3.10, no separate `org.jetbrains.kotlin.android` plugin), Gradle 9.4.1, KSP 2.3.9. The Gradle wrapper (`gradlew`, `gradlew.bat`, `gradle-wrapper.jar`, and `gradle-wrapper.properties` with its distribution SHA-256 checksum) is committed, so a fresh clone can build without a pre-existing local Gradle install. An earlier Android baseline (predating the ongoing-notification milestone) was successfully validated with a real local run — `assembleDebug` produced a debug APK, `lintDebug` completed with 0 errors, and `testDebugUnitTest` passed 117 JVM unit tests — on a machine with JDK 17 and the Android SDK. The notification-foundation implementation that followed, with 193 JVM source-level `@Test` functions, was itself later validated with a fresh local `./gradlew testDebugUnitTest lintDebug assembleDebug` run: `testDebugUnitTest` passed with no failures, `lintDebug` completed with 0 errors/37 warnings, and `assembleDebug` produced a debug APK. (This run also caught and fixed two real issues surfaced only by actually executing the notification code: a test assertion that could false-fail against a departure's own line designation rather than an actual countdown value, and two Android Lint findings — a `MissingPermission` false positive on the already-guarded notification post, and two `context.getString()` calls inside non-composable callbacks that Lint flagged for potential staleness across configuration changes, both resolved by resolving the strings once via `stringResource()` in composable scope.)

The scheduling/active-window/permission-flow milestone added 42 further JVM `@Test` functions and 3 further instrumented Compose UI `@Test` functions beyond the 193-test baseline (reaching 235 JVM / 9 instrumented in source at the time); a subsequent corrective audit session fixing the device-local-timezone bug, the shared notification-availability check, the lifecycle-aware status refresh, and the explicit one-routine dialog added a further 23 JVM `@Test` functions and 3 further instrumented `@Test` functions, reaching 258 JVM / 12 instrumented. All of that was subsequently compiled and run for the first time, together with a further session's `BOOT_COMPLETED` receiver and durable stale-snapshot storage (adding 2 JVM and 9 instrumented tests), reaching 260 JVM / 21 instrumented. The promoted Live Update request, its Stop action, and the pause-today device-timezone fix added 10 further JVM `@Test` functions with no further instrumented ones, reaching 270 JVM / 21 instrumented. The active-window worker's notification-availability re-check on every loop tick added 1 further JVM `@Test` function with no further instrumented ones, reaching 271 JVM / 21 instrumented. The corrected debug promotion-status wording, the Settings deep-link, the About screen, and the forecast-based direction-discovery fix added 4 further JVM `@Test` functions and 3 further instrumented ones, reaching 275 JVM / 24 instrumented. A correction pass fixing that deep-link's missing Android-16+ gate (`shouldOfferLiveUpdateSettingsLink`) added 3 further JVM `@Test` functions with no further instrumented ones, reaching 278 JVM / 24 instrumented. A second correction pass, fixing the settings link's remaining silent-failure fallback case (`launchLiveUpdateSettings`), added 2 further JVM `@Test` functions with no further instrumented ones, reaching 280 JVM / 24 instrumented. The home-screen widget implementation (`RoutineWidgetMapper`, `RoutineWidgetUpdater`, `BlickRoutineWidget`) added 53 further JVM `@Test`



functions with no further instrumented ones, reaching 333 JVM / 24 instrumented. A final re-audit session fixing that widget's lifecycle/visual gaps (Stop-action wiring, every worker exit path, the `NotificationsUnavailable` state, the zero-delay reschedule fix, the freshly-placed-widget reconcile fix, responsive sizing, and the readable theme background) added 8 further JVM `@Test` functions with no further instrumented ones, reaching 341 JVM / 24 instrumented. A "Design 1" visual redesign session (the colored line-number badge via `LineBadgeColorMapping`, the large-countdown layout, and the width-driven responsive size tiers) added 23 further JVM `@Test` functions with no further instrumented ones — **364 JVM `@Test` functions and 24 instrumented `@Test` functions now exist in source, and all of them pass.** See [android/README.md](#)'s Build section for the exact toolchain versions, the up-to-date per-file test breakdown, and the real issues that first full run found and fixed (including a genuine production bug, not just test-suite corrections), and see "Validation status of this update" at the top of this document for the summary.

## Backend

The backend uses:

- TypeScript;
- Hono;
- Zod validation;
- a Vercel-compatible handler;
- a portable application core shared with a local Node development server;
- explicit upstream-client, normalization, cache, and error-handling interfaces;
- Vitest for backend tests.

The backend remains necessary even though the chosen upstream APIs are keyless. It provides:

- a stable, versioned contract for Android and a future iPhone client;
- upstream schema isolation;
- validation and normalization;
- caching and request deduplication;
- fair-use protection;
- consistent timestamps and errors;
- a controlled place to adapt when upstream services change.

No Trafiklab API key is required by the selected endpoints. `.env.example` should therefore contain only genuinely required runtime configuration, such as local port or environment mode.

## Platform-neutral data flow

| Stage                      | Responsibility                                                                                        |
|----------------------------|-------------------------------------------------------------------------------------------------------|
| Android now / iPhone later | Stores routines, schedules active periods, calculates countdowns, and presents information            |
| Blick backend              | Validates requests, fetches and normalizes upstream data, applies caching, and returns versioned JSON |
| SL Transport               | Supplies sites, departures, line IDs, direction codes, stop data, and embedded deviation signals      |
| SL Deviations              | Supplies detailed and time-bounded disruption messages                                                |

The backend must not return Android-specific notification, Room, or UI concepts.

---

## 11. Backend API contract

All routes are versioned under `/api/v1`.

### Envelopes

Successful response:

```
{
  "schemaVersion": 1,
  "data": {}
}
```

Error response:

```
{
  "schemaVersion": 1,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "A safe, actionable message"
  }
}
```

Supported machine-readable error codes:

```
VALIDATION_ERROR
UPSTREAM_ERROR
UPSTREAM_TIMEOUT
UPSTREAM_RATE_LIMITED
NOT_FOUND
RATE_LIMITED
INTERNAL_ERROR
```

Errors must not expose stack traces, environment values, or unnecessary upstream details.

### Health

`GET /api/v1/health`

Returns service status and a backend timestamp.

Cache policy:

Cache-Control: no-store

### Site search

`GET /api/v1/stops/search?query=`

The backend searches a cached snapshot from SL Transport `/v1/sites?expand=true`.

Normalized site:

```
siteId
name
note
lat
lon
stopAreaIds
```

Search requirements:

- trim and validate the query;
- use case-insensitive and diacritic-tolerant matching;
- search site name and nullable note;
- rank exact, prefix, token-prefix, and substring matches in that order;
- use deterministic tie-breaking;
- return no more than 20 results.

Cache policy:

Cache-Control: public, s-maxage=3600, stale-while-revalidate=86400

## Departures

GET /api/v1/departures?siteId=&forecast=

`siteId` is required and must be a valid positive SL Site ID. `forecast` is optional: a positive integer number of minutes to look ahead, capped at 1200 (20 hours) — an empirically observed point past which the real upstream silently returns zero departures rather than erroring, not a maximum documented or guaranteed by Trafiklab/SL (see [docs/api-contract.md §3](#)). When omitted, the upstream's own short undocumented default applies, as before. Android's routine-setup direction discovery passes `forecast=1200` specifically; the live routine-details and notification polling paths never pass it.

Top-level data:

```
fetcherAt
timeZone
siteId
departures
siteDeviations
```

`timeZone` is always `Europe/Stockholm`. `fetcherAt` records when the backend actually fetched the upstream response, not when a cached response was later served.

Normalized departure:

```
departureId
journey
line
direction
directionCode
destination
via
stopArea
stopPoint
scheduledTime
expectedTime
state
isCancelled
tripDeviations
```

The defensive departure identity is:

`journey.id + ":" + stopPoint.id + ":" + scheduledTime`

`expectedTime` is nullable. Clients use `expectedTime` when available and otherwise use `scheduledTime`. The backend never returns `minutesRemaining`; each client calculates it from the effective timestamp and the current device time.

`isCancelled` is true when:

- the departure state is `CANCELLED`; or
- a trip deviation has consequence `CANCELLED`.

Other unfamiliar state strings remain available for forward compatibility and are not reinterpreted as cancellation.

Cache policy:

Cache-Control: public, s-maxage=30, stale-while-revalidate=30

## Disruptions

GET /api/v1/disruptions?siteId=&lineId=&transportMode=&future=

`siteId` is required. The remaining filters are optional and validated. `future` defaults to `false`, which limits the normal active-routine request to currently published disruptions. A caller must opt in explicitly when future disruption messages are needed.

Normalized disruption:

```
disruptionId
version
createdAt
modifiedAt
validFrom
validUntil
priority
message
affectedStopAreas
affectedLines
affectedModes
```

`priority` contains the upstream importance, influence, and urgency levels. These values are sorting hints and are not renamed to `severity`.

`message` contains:

```
header
details
scopeAlias
webLink
language
```

The backend selects the message variant whose language is `sv`. It falls back to the first available variant only when no Swedish variant exists.

`affectedModes` is a documented derived list of unique transport modes from affected lines.

Cache policy:

Cache-Control: public, s-maxage=60, stale-while-revalidate=60

The public contract does not include an unused lines endpoint. One may be added later only when it has a defined consumer, normalized schema, and tests.

---

## 12. Local information model

### Commute routine

A Room `RoutineEntity` supports multiple records even though the first UI exposes one:

```
id
name
siteId
siteName
transportMode
lineId
lineDesignation
directionCode
destinationLabel
activeDaysMask
startTimeMinutes
endTimeMinutes
enabled
pausedDateEpochDay
```

`siteId`, `lineId`, `transportMode`, and `directionCode` are the stable matching fields. `destinationLabel` is a presentation value and must not be the sole identity — there is no separate `directionLabel` field; direction is represented only by `directionCode`. `activeDaysMask` stores the active weekdays as a bitmask (bit 0 = Monday ... bit 6 = Sunday) rather than a list, and `startTimeMinutes`/`endTimeMinutes` store the active window as minutes-since-midnight. `pausedDateEpochDay` is nullable and holds an epoch-day value only when the routine is paused for a specific date. The entity does not track `createdAt`/`updatedAt` timestamps — routines are identified and ordered by their own fields (currently `name`), not by creation history.

The Room DAO exposes:

```
Flow<List<RoutineEntity>>
getById
upsert
update
delete
deleteById
```

`upsert` is `@Insert(onConflict = OnConflictStrategy.REPLACE)` — an insert that replaces an existing row sharing the same primary key, rather than a separate merge/patch operation.

Preferences DataStore holds only small application settings such as first-launch state, theme choice, and whether an explanation has been dismissed.

## Client domain timestamps

Network DTOs receive ISO 8601 strings with explicit offsets. Android maps them into `java.time.Instant` or `ZonedDateTime` domain values instead of retaining raw strings. A future iPhone client will perform an equivalent native conversion.

---

## 13. Active-window scheduling and the 30-second notification loop

*Implemented in source and verified — see "Current implementation status" above and the "Validation status" note at the top of this document for the full local run, including manual exercising of the active-window scheduling/posting/removal loop on a physical device.*

Android limits background activity to protect battery life. Blick does not attempt to run continuously — it schedules work only around each routine's own configured window.

### Why WorkManager, and why not an exact alarm

- `PeriodicWorkRequest` was **not** used for the 30-second in-window refresh: WorkManager's periodic work has a 15-minute minimum interval, which cannot represent a 30-second cadence. Instead, a single `OneTimeWorkRequest` per routine activates the next window

(`WorkManagerRoutineScheduler.scheduleActivation`), and once that worker starts, it owns its own internal 30-second loop for the duration of that one window.

- `USE_EXACT_ALARM` and `SCHEDULE_EXACT_ALARM` are deliberately **not** used, and there is no exact-alarm permission screen. Start-time activation is therefore best-effort, not exact — WorkManager does not guarantee the worker starts at precisely the configured time, only that it will start at or after it, subject to normal Android battery/Doze/OEM scheduling behavior. This document and the app must not claim exact execution.
- Once activated, the worker calls `setForeground()` with a `CoroutineWorker`, so WorkManager itself manages promoting the work to a foreground service — no separate, manually-started foreground service class was written. See [Long-running workers](#).
- The `dataSync` foreground-service type was chosen (`FOREGROUND_SERVICE_DATA_SYNC` permission, `android:foregroundServiceType="dataSync"` on the merged `SystemForegroundService` manifest entry) since the worker's job — fetching current departures over the network and reflecting them in a notification — is a data-sync operation, not media, location, or camera/mic use. See [Foreground service types](#). Android 15+ limits a `dataSync` foreground service to six total hours in any rolling 24-hour period — routine windows are expected to be short (a typical commute window), which is why the worker stops reliably at the routine's configured end time rather than running indefinitely.

## Scheduling, replacing, and cancelling work

Each routine's scheduled activation uses a unique, replaceable WorkManager work name (`ExistingWorkPolicy.REPLACE` against `"routine-active-window-" + routineId`). Saving a new routine, editing an existing one, enabling, disabling, pausing for today, resuming, or deleting a routine all call through the same `RoutineScheduler.scheduleActivation / cancelActivation` interface, so a change can never leave stale, obsolete scheduled work behind for that routine. `NextOccurrenceCalculator` computes the next active window (or "active right now") from the routine's active weekdays and start/end time using `ZonedDateTime`, so the JDK itself resolves daylight-saving-time gaps and overlaps rather than the app guessing. The current instant (an injectable `Clock`) and the zone used to interpret it (an injectable `DeviceZoneProvider`, resolving `ZoneId.systemDefault()` fresh on every call in production) are deliberately separate: a routine's weekdays and start/end time are the device's own local wall-clock values, so combining a zone-less `Clock.instant()` with the device's current zone is what makes a 07:30 Stockholm routine actually activate at 07:30 Stockholm time, in both directions of daylight saving, rather than being silently reinterpreted against the clock's own zone (e.g. UTC). Because the zone is re-resolved on every call rather than captured once, a live device timezone change is picked up automatically the next time a schedule is computed, and is also reconciled immediately via a runtime-registered `ACTION_TIMEZONE_CHANGED` receiver.

## The 30-second loop itself

Once a window is active, `RoutineActiveWindowWorker.doWork()`:

- 1 Loads and validates the routine, and checks whether it has been started late (for example after a delay imposed by the system); if the window has already elapsed, it skips posting entirely and reschedules the next eligible occurrence rather than showing a notification for a window that has already ended.
- 2 Checks shared notification availability (the same `NotificationAvailabilityChecker` the notifier and the Routine Details status hint use) **before** ever entering foreground execution. If notifications are unavailable — permission missing, app notifications disabled, or the Blink channel specifically disabled — the worker never calls `setForeground()`, never starts the loop, never recreates or modifies a disabled channel, and never reports delivery as active; it simply reschedules the next eligible occurrence and exits. Only once availability is confirmed does it enter foreground execution with a valid Blink notification (channel creation still proceeds normally if the channel is merely missing rather than disabled).

- 3 Fetches current departures through the existing live-departures engine.
- 4 Filters and maps them through the existing notification model (up to two departures), and posts or silently updates the routine's one ongoing notification — same notification id every time, `setOnlyAlertOnce(true)`, no repeated sound, vibration, heads-up, or extra card — requesting promotion to a lock-screen Live Update on every post (see "Requesting a promoted Live Update" below).
- 5 Waits about 30 seconds.
- 6 Repeats from step 3 until the routine's configured end time, a disable, or a deletion — whichever happens first (re-reading the routine from storage on every iteration so an edit made mid-window takes effect without waiting for the next scheduled activation).
- 7 Removes that notification, stops, and schedules the next eligible occurrence.

If a network fetch fails after an earlier successful one inside the same window, the existing stale/offline semantics apply (no extra alert; the same notification reflects the last known departures with an appropriate stale indication) and the next 30-second tick retries. That stale snapshot is Room-backed (`StaleSnapshotRepository`), keyed by routine id and scoped to the exact departure identity that produced it, and shared with the routine details screen — it survives process death rather than being scoped to one worker run, and either the worker's or the screen's own successful fetch can serve as the other's fallback.

Terminal handling is deliberately three-way: normal completion, a handled failure (for example building the foreground notification or posting an update throws), and a late start or unavailable notifications all re-read the routine and reschedule its next eligible occurrence (unless the routine has since been deleted or disabled) and clean up the active notification if the loop had actually been entered. A genuine `CancellationException` — from the routine being deleted, disabled, or its unique work being replaced mid-run — is always rethrown rather than caught as a handled failure, so cancellation never gets converted into a success/failure result and never causes the cancelled work to be rescheduled or resurrected.

## Requesting a promoted Live Update

Every post or silent update of the routine's ongoing notification also requests promotion to Android 16's Live Update surface, rather than settling for an ordinary notification-drawer entry:

- `RoutineNotificationBuilder` calls `setRequestPromotedOngoing(true)` on the `NotificationCompat.Builder` for every content state, and declares the manifest permission `android.permission.POST_PROMOTED_NOTIFICATIONS` (normal, granted at install time, distinct from the runtime `POST_NOTIFICATIONS` permission).
- The notification body uses `BigTextStyle` (listing both visible departures) with `setShortCriticalText` set to the soonest departure's countdown, or a cancelled indicator — one of the styles the promoted-notification surface actually supports. No custom `RemoteViews` are used anywhere in the builder, since promoted Live Updates do not support them.
- `PromotedNotificationChecker` (backed by `NotificationManagerCompat.canPostPromotedNotifications()`) reports whether the OS currently *permits requesting* promotion — eligibility, not confirmation that any specific surface (e.g. Samsung's Now Bar) actually rendered a card, which no API exposes to a third-party app. The Routine Details debug section's wording was corrected to state this plainly, after an earlier version of this document overclaimed it as confirming the Live Update had appeared. It is not used to change what gets posted — the same builder call is made either way, so an unsupported device or a user who has disabled promoted notifications simply receives a standard ongoing notification automatically, with no separate fallback code path to keep in sync. When notifications are available but promotion currently isn't eligible, the routine details screen also offers a link to Android's own per-app Live Update settings (`Settings.ACTION_APP_NOTIFICATION_PROMOTION_SETTINGS`), shown only on Android 16+; on an OEM build that omits the screen even there, launching it falls back to the ordinary per-app notification settings screen instead of leaving the tap do nothing — see below.



- Because `setRequestPromotedOngoing` was only stabilized in `androidx.core` 1.17.0, and the next stable release (1.19.0) requires `compileSdk` 37 (one major version beyond this project's current 36), the dependency is deliberately held at 1.17.0 rather than the latest available version — see `android/README.md`'s Build section.
- **On a real Android 16 device (Samsung Galaxy S23 Ultra), the promoted card did not appear by default — only after manually enabling Settings → Developer options → "Live notifications for all apps."** Blick's own implementation is standard and correct; this restriction is entirely Samsung's, not Blick's, and not fixable in app code. The official Android docs explicitly note that "OEMs can enforce additional criteria for Live update eligibility," and current, independent reporting confirms third-party Now Bar access remains restricted on **both One UI 8 and One UI 8.5** (this device's actual build, not a beta) unless that developer flag is manually enabled. **An earlier version of this document claimed the restriction was beta-only and would lift once One UI 8 shipped stable — that was tech-press speculation, not an official Samsung statement, and this device (already on 8.5) still requiring the flag disproves it. No official Samsung source for a removal date has been found; treat the Now Bar experience as device/firmware-dependent, not a guaranteed feature, and do not restate a resolution timeline without a real source.** There is no manifest flag, permission, or API call available to a third-party app that enables Samsung's developer option — Blick cannot bypass it — and **ordinary users must never be instructed to enable Developer options** as a workaround. With the flag off (the default for any ordinary user), the app's `setRequestPromotedOngoing(true)` request is simply not honored: the OS posts a plain ongoing notification instead, with no error and nothing in `canPostPromotedNotifications()`'s result visibly distinguishing it from the designed fallback path — this project's debug promotion-status line has not been separately confirmed to reflect Samsung's own gate one way or the other. Once the developer flag was enabled, the promoted card appeared on the lock screen near the routine's configured start time, with the same behavior otherwise exercised through the regular-notification testing below (single notification, ~30-second refresh, up to two departures, survives Blick being swiped from Recent Apps, disappears at the routine's end time). Separately, Android's own docs describe a real, permanent, user-facing settings control for the platform's own Live Update eligibility — `Settings.ACTION_APP_NOTIFICATION_PROMOTION_SETTINGS` — the routine details screen now deep-links to it, shown only on Android 16+, falling back to the ordinary per-app notification settings screen on an OEM build that omits it even there; this is a distinct, general Android control that **cannot enable Samsung's separate "Live notifications for all apps" developer option**, so it does not make Blick's Now Bar card generally available on Samsung devices either way. This project's Android 14 physical test device (Lenovo TB350FU) cannot show the promoted surface at all regardless, since it's an Android 16-only platform feature — that device correctly falls back to a plain ongoing notification instead, the same fallback a regular Samsung Android 16 user currently gets by default.
- Separately from promotion specifically, the full notification/scheduling loop was manually verified end to end on the Samsung Galaxy S23 Ultra in its default (non-promoted) state — the exact experience a regular user has today: exactly one notification is ever shown; it refreshes about every 30 seconds with up to two departures; it appears on the lock screen; it survives Blick being swiped away from Recent Apps; it disappears at the routine's configured end time; disabling an active routine removes its notification and re-enabling one during its active window restores it; reboot recovery (`BootCompletedReceiver`) still works; and tapping the notification while Blick is already open does not create a duplicate screen. This is the coverage that matters for every real user until promotion's rollout gate (above) opens up.
- The notification always carries a Stop action (a plain `NotificationCompat.Action`, not a custom view, so it stays valid on the promoted surface too) fulfilling the spec's "Stop/Unpin" requirement. Its `PendingIntent` targets `StopRoutineNotificationReceiver` — an `exported="false" @AndroidEntryPoint BroadcastReceiver` only ever triggered by this app's own explicit intent, never a system or cross-app broadcast — which hands off to `StopRoutineNotificationAction`, a small, separately unit-tested class (the same split used for `BootCompletedReceiver` and `RoutineScheduleReconciler`). Stopping today's window early is given exactly

the same effect as the existing "pause for today" control: it writes `pausedDate` to today's date and reschedules, which `RoutineActiveWindowWorker`'s own next loop tick already observes and stops on (see "The 30-second loop itself" above) — and the notification is also removed directly, so it disappears immediately on tap rather than up to ~30 seconds later. "Today" here is deliberately resolved from the device's current zone (mirroring the worker's own `zonedNow()`), not a zone-less clock, so this write and the worker's read of it always agree, including right around local midnight. Introducing this surfaced a pre-existing bug in `RoutineDetailsViewModel`'s own "pause today"/"resume today" controls, which had been resolving "today" from a zone-less `LocalDate.now(clock)` — the production `Clock` is `Clock.systemUTC()` (see `di/TimeModule.kt`), so shortly after local midnight in any zone ahead of UTC (e.g. Sweden), "pause today" could pause yesterday instead, disagreeing with the worker's own device-zone break condition. Fixed to resolve "today" the same `DeviceZoneProvider`-based way everywhere in this screen. Verified on the same real Samsung Galaxy S23 Ultra: Stop removes the notification immediately and marks the routine "Paused today"; the notification does not return while paused, including when Stop is tapped directly from the locked screen; and Resume today both clears the pause and restarts the notification immediately if the routine's window is still active.

## Reboot and process death

WorkManager persists scheduled work through ordinary process death and device reboot on its own. `BlickApplication.onCreate()` additionally reschedules every enabled saved routine at process start, as an idempotent reconciliation safety net — this is not the primary scheduling path. A dedicated `BootCompletedReceiver` runs that same reconciliation directly on `ACTION_BOOT_COMPLETED`, as a further backstop for a reboot after which the app process never happens to start on its own before a routine's next window would have opened; it only ever enqueues WorkManager work from that receiver and does not start a foreground service directly from `BOOT_COMPLETED` (restricted on modern Android; see [Restrictions on background starts](#)).

## What remains best-effort, not exact

- Start-time activation is best-effort, as stated above — this is a deliberate, documented trade-off, not an oversight.
- The 30-second refresh cadence inside an active window is enforced by the worker's own `delay()` loop while it holds foreground execution, not by a system-guaranteed timer.

Relevant Android documentation:

- [Long-running workers](#)
- [Foreground service types](#)
- [Restrictions on background starts](#)
- [Background work](#)
- [Create notifications](#)

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## 14. Time, caching, and resilience

### Stockholm timestamps

SL Transport emits local Stockholm timestamps without an explicit UTC offset. The backend interprets them using the IANA zone `Europe/Stockholm` and emits ISO 8601 timestamps with an explicit offset.

The conversion must not silently rely on a date library's default daylight-saving-time choice. Tests cover:

- normal winter and summer timestamps;
- both sides of the spring 2026 transition;

- both sides of the autumn 2026 transition;
- the duplicated local hour during the autumn overlap;
- a nonexistent local time during the spring gap.

For an ambiguous autumn timestamp, the backend uses the upstream fetch time and forecast window to select the chronologically plausible occurrence. A malformed or impossible timestamp is handled explicitly rather than silently shifted.

SL Deviations already emits timestamps with offsets; they are validated and preserved.

## Serverless caching

The backend uses a [Cache](#) interface with an in-memory implementation for local development and tests, and for best-effort per-instance reuse of the site-directory snapshot (which is not subject to a per-minute fair-use limit).

Important limitations and, for SL Deviations specifically, how they are now closed:

- Vercel serverless instances do not share dependable process memory;
- a daily site snapshot in memory is not a guaranteed global daily cache — this is fine for site data, which changes at most once per day per SL Transport's own docs;
- HTTP cache headers are the dependable initial shared layer for public GET responses;
- simultaneous identical requests within one instance should be deduplicated;
- **the SL Deviations one-request-per-minute limit is now enforced in production, closing what was previously a blocker for public deployment.** The two compounding gaps that used to make it unenforceable — per-instance-only cache/dedup, and a cache keyed per query combination rather than per upstream overall — are both closed by fetching exactly one shared, network-wide SL Deviations snapshot (never one call per query) and coordinating every Vercel instance's access to it through a Redis-backed (Upstash) distributed lock and cache, with the 60-second floor covering failed attempts too and a 6-hour stale-fallback snapshot for when a refresh does fail. See [docs/api-contract.md](#), "Caching and fair use", for the full design, and [backend/README.md](#), "Redis (Upstash) setup", for the required production configuration (`UPSTASH_REDIS_REST_URL/UPSTASH_REDIS_REST_TOKEN` — the backend refuses to start in production without them, never silently falling back to the per-instance-only implementation).

## Stale and unavailable information

When live information cannot be refreshed, the application must not present old data as current. Appropriate states include:

```
Unable to update - last checked 07:42
Scheduled time only
No upcoming departures
```

The backend applies timeouts and consistent error mapping. The Android client retains the last successful response only with a visible stale state and appropriate expiry.

The retained response is scoped to the exact departure identity (site, line, direction, and transport mode) that produced it. If the user edits a routine to a different site/line/direction/mode and that new configuration's first fetch fails, the client must not fall back to the previous configuration's retained response — doing so would mislabel unrelated departures as "stale" data for the new routine. A failed refresh may only fall back to stale data that was captured for the identical configuration currently being fetched.

---

## 15. Edge cases

The application must account for:

- no upcoming departures;
  - expected time missing while scheduled time is available;
  - temporarily unavailable real-time information;
  - a cancelled departure;
  - an unfamiliar future state value;
  - a changed destination;
  - a changed stop point or platform;
  - several stop positions at one site;
  - a shortened route;
  - a line-wide or mode-wide disruption;
  - a disruption without a Swedish message variant;
  - the device being offline;
  - the phone being restarted;
  - notification permission being denied;
  - lock-screen content being hidden;
  - Android delaying scheduled work;
  - a routine spanning midnight;
  - daylight-saving-time changes;
  - weekday and holiday timetable differences;
  - upstream timeouts, malformed data, rate limiting, or service failure;
  - a route not appearing during setup because it is outside the current departure forecast.
- 

## 16. Minimum viable product

The MVP contains:

- 1 Native Android setup application.
- 2 One routine exposed in the UI.
- 3 Room persistence designed for multiple routines.
- 4 SL site search.
- 5 Bus, metro, commuter-train, and local-rail support.
- 6 Transport, line, and direction selection from available data.
- 7 Weekday selection.
- 8 Configurable start and end time.
- 9 Up to two departures total when available.
- 10 One scheduled ongoing notification.
- 11 Lock-screen visibility subject to Android settings.
- 12 Relevant SL disruption messages.
- 13 Cancellation presentation.
- 14 Automatic removal at the end of the routine.

- 15 Pause-for-today control.
- 16 Stale and offline states.
- 17 Visible Trafiklab attribution.
- 18 Versioned Hono backend deployed to Vercel.

Outside the MVP:

- user accounts;
  - cloud routine synchronization;
  - GPS tracking;
  - analytics;
  - advertising;
  - social features;
  - arbitrary journey planning;
  - iPhone implementation;
  - TV implementation;
  - casting and device pairing;
  - all-Sweden operator coverage;
  - walking-time advice;
  - predictive "Can I catch it?" messages;
  - aggressive alert sounds;
  - smartwatch support.
- 

## 17. Planned and possible later versions

### Planned: native iPhone client

A native iPhone application is planned after the Android MVP. It will consume the same versioned backend contract and reuse the documented transport identity, filtering, timestamp, and cancellation rules.

The iPhone phase must separately investigate:

- Swift and SwiftUI implementation;
- ActivityKit and Live Activities;
- Apple Push Notification service;
- iOS background-execution constraints;
- local persistence;
- App Store delivery and testing.

The Android notification design must not be assumed to translate directly to iOS.

### Possible: Smart TV departure display

A Smart TV display is a possible future feature. The preferred first investigation is a secure web or casting experience using the existing backend, rather than separate applications for every TV operating system.

Potential requirements include:

- a large, remote-friendly departure board;

- secure pairing or a temporary display link;
- explicit consent before sharing a saved routine;
- cross-device configuration without exposing travel routines.

No TV, casting, pairing, or cloud-synchronization code is included now.

### Other possible additions

- multiple visible routines;
- separate morning and evening schedules;
- holiday and temporary schedule controls;
- improved direction discovery;
- accessibility and theme options;
- Wear OS investigation;
- additional Swedish operators after a separate data-contract review.

---

## 18. Privacy and security

The first Android application does not require the user's identity or precise location.

MVP privacy choices:

- no account;
- routines stored locally in Room;
- no GPS permission;
- no analytics;
- no advertising identifiers;
- no travel-history profile;
- no cloud synchronization;
- no cross-device sharing.

The backend receives public SL identifiers and filters needed for each request. A saved site and routine can still reveal a travel pattern, so logs must avoid retaining complete routine behaviour or associating requests with an identifiable person unnecessarily.

Security requirements:

- HTTPS only;
- strict query validation;
- safe, consistent errors;
- upstream timeouts;
- request-rate protection;
- no stack traces or configuration details in responses;
- environment-based configuration;
- dependencies kept current;
- platform-neutral contract tests.

Although the current upstream APIs are keyless, the backend must remain capable of protecting credentials if a future upstream service requires them.

---

## 19. Testing and release requirements

### Backend tests

- request validation and error envelopes;
- site-search ranking, Swedish characters, and result limits;
- upstream-to-normalized serialization;
- departure identity;
- nullable expected time;
- cancellation from state and trip deviation;
- unknown state preservation;
- Swedish disruption-message selection and fallback;
- cache expiry and in-flight deduplication;
- timeout and upstream failure mapping;
- Stockholm daylight-saving-time transitions.

### Android tests

- Room insert, update, delete, and observation;
- routine enable, disable, and pause behaviour;
- DTO-to-domain timestamp conversion;
- effective-time and countdown calculation;
- filtering by site, line, mode, and direction code;
- ViewModel behaviour where implemented;
- stale and offline presentation;
- notification presentation mapping — a pure, Android-free mapper from a routine and the live-departures engine's state to a notification model (`RoutineNotificationMapperTest`);
- notification construction and posting — the real, built `android.app.Notification` and the real framework `NotificationManager`, exercised via Robolectric (channel importance and idempotency, ongoing/visibility/icon flags, tap-intent routine id, per-state content, the app-wide notifications-disabled case, and the Blink notification channel's own `IMPORTANCE_NONE` case — distinct from the app-wide toggle);
- notification-tap intent consumed exactly once, so an Activity recreation cannot replay an already-handled tap;
- `RoutineDetailsViewModel`'s debug notification trigger, via hand-written fakes.

The tests above, and the tests below, have all since been executed via a fresh local `./gradlew testDebugUnitTest lintDebug assembleDebug connectedDebugAndroidTest` run, with no failures — see "Validation status of this update" at the top of this document.

- Add-routine control: an instrumented Compose UI test (`RoutineListScreenTest`, calling the extracted stateless `RoutineListContent` composable directly, no Hilt needed) proving the FAB exists and opens the creation flow both when the routine list is empty and when it already has a saved routine, and that tapping a saved routine still opens it.
- `RoutineDetailsViewModel`'s 30-second auto-refresh: immediate fetch on first becoming active without a duplicate fetch (since `init` already fetched once), fetch again after ~30 virtual seconds, no `Loading` flash on an automatic tick, cancellation stops further ticks when the screen becomes inactive, an immediate fetch on



reactivation, and no duplicate/overlapping refresh loop across repeated activation — all using the existing `StandardTestDispatcher`/virtual-time convention, never a real 30-second wait.

- `NextOccurrenceCalculator`: no active weekday, before/inside/at the boundary of today's window, rolling forward across inactive weekdays and into the following week, the paused-for-today exclusion, every day active, and both sides of the 2027 spring daylight-saving gap and the 2026 autumn overlap.
- `WorkManagerRoutineScheduler`: real `WorkManager` (via `WorkManagerTestInitHelper.initializeTestWorkManager`, not a hand-rolled fake) enqueue, replace-on-reschedule, and cancel behavior for a routine's unique work name.
- `RoutineActiveWindowWorker` (via `TestListenableWorkerBuilder` with a fake worker factory): missing routine id fails cleanly; a routine that no longer exists succeeds without posting; a late start past the window's end skips posting and reschedules; a normal run produces the correct number of 30-second ticks, calls `notifier.remove()` exactly once at the end, and reschedules once; the notification model always carries the correct routine id for tap navigation; the routine being disabled mid-run stops the loop early and reschedules with the routine's new disabled state; the routine being deleted mid-run stops the loop and does **not** reschedule; and a fetch failure after an earlier success in the same window produces a `Stale` notification content, not `Offline/Unavailable`.
- Scheduler integration: enabling, disabling, pausing for today, resuming, and deleting a routine from `RoutineDetailsViewModel`, saving a new routine from `RoutineCreateViewModel`, and deleting a routine from `RoutineListViewModel` all call through to a fake `RoutineScheduler`'s `scheduleActivation/cancelActivation` as expected.

## Release checks

- Android build, lint, and unit tests;
  - backend type-check, lint, tests, and production build;
  - Vercel health and cache-header checks;
  - real-device notification and lock-screen testing;
  - scheduling tests across reboot and battery restrictions;
  - attribution and terms review;
  - no secrets or local environment files committed;
  - no unrelated screenshots or generated build output in Git.
- 

## 20. Success criteria

The MVP is successful when a user can configure a routine once and then, on subsequent selected weekdays:

- see the correct site, line, and direction automatically;
- see current upcoming departures in one Android notification;
- see the next departure replace the previous one;
- see relevant disruptions without unrelated messages;
- see cancellations and stale-data states clearly;
- have the notification disappear when the active period ends;
- use the feature without reopening the application each morning.

Technical success includes:

- reliable scheduled activation on supported Android devices;

- reasonable battery use;
  - correct timestamp and daylight-saving-time behaviour;
  - stable platform-neutral backend responses;
  - compliance with upstream fair-use and attribution requirements;
  - no unnecessary collection of personal information;
  - no dependency on an exposed mobile API key.
- 

## 21. Short project description

**Blick is an Android-first scheduled departure display for regular SL commuters. Users choose a site, transport mode, line, direction, weekdays, and time window. During that period, upcoming departures — and relevant disruptions, in both the Routine Details screen and the notification — appear automatically in one calm, updating Android notification (see "Current implementation status" above). The platform-neutral backend is designed to support a planned native iPhone client and possible future household displays without expanding the initial MVP.**